

YINGKOU, China

WMO: 544710

Lat: 40.6656N

Lon: 122.1733E

Elev: 4

StdP: 101.28

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.5	-15.4	-26.2	0.3	-13.6	-23.9	0.4	-12.1	10.0	-3.4	8.9	-6.2	2.4	40	0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.5	30.6	24.4	29.6	23.9	28.6	23.4	26.3	29.1	25.5	28.2	24.8	27.5	4.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.5	20.8	28.4	24.6	19.7	27.6	23.9	18.8	26.9	82.5	29.2	78.8	28.4	75.7	27.5	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 9.7	8.5	7.5	DB	-19.1	32.4	3.3	1.3	-21.5	33.3	-23.4	34.1	-25.2	34.8	-27.6	35.7	(4)
(5)			WB	-19.7	27.1	3.1	0.9	-21.9	27.8	-23.8	28.3	-25.5	28.8	-27.7	29.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	-7.9	-3.9	2.9	11.1	17.6	22.3	25.5	24.7	19.7	11.9	2.5	-5.0	(6)
(7)		DBStd	12.18	4.40	4.45	4.15	3.76	2.92	2.29	1.86	2.25	3.05	4.33	5.37	4.61	(7)
(8)		HDD10.0	1922	554	388	223	31	0	0	0	0	0	31	229	465	(8)
(9)		HDD18.3	3602	813	622	480	219	48	2	0	0	21	201	474	723	(9)
(10)		CDD10.0	1996	0	0	2	63	236	371	479	457	291	91	6	0	(10)
(11)		CDD18.3	633	0	0	0	1	26	122	221	199	62	2	0	0	(11)
(12)		CDH23.3	4295	0	0	0	3	91	577	1792	1564	267	2	0	0	(12)
(13)		CDH26.7	872	0	0	0	0	9	86	409	343	24	0	0	0	(13)
(14)	Wind	WSAvg	3.5	3.1	3.4	4.1	4.5	4.1	3.5	3.1	3.0	3.1	3.5	3.6	3.2	(14)
(15)	Precipitation	PrecAvg	675	7	8	12	39	53	73	182	159	76	41	18	9	(15)
(16)		PrecMax	1116	31	30	32	130	163	193	553	428	133	135	56	39	(16)
(17)		PrecMin	387	0	0	0	4	16	15	35	16	2	0	0	0	(17)
(18)		PrecStd	166	7	8	9	30	33	43	111	100	34	32	14	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.0	8.0	15.6	23.0	27.9	30.6	32.4	31.6	28.9	23.1	16.4	8.3	(19)
(20)			MCWB	1.3	3.0	9.1	14.0	17.4	20.7	25.2	25.4	22.0	17.9	12.0	5.9	(20)
(21)		2%	DB	2.2	6.2	12.8	20.4	25.6	28.8	31.1	30.3	27.2	21.4	14.4	5.7	(21)
(22)			MCWB	-0.3	2.3	7.2	12.2	16.5	20.6	24.9	24.5	20.8	16.6	11.0	3.4	(22)
(23)		5%	DB	0.7	4.5	10.8	18.6	24.1	27.5	29.9	29.4	26.0	19.9	12.8	3.8	(23)
(24)			MCWB	-1.5	1.1	5.8	11.5	15.8	20.3	24.6	24.2	20.1	15.6	9.7	1.7	(24)
(25)		10%	DB	-0.8	2.9	9.0	17.0	22.7	26.3	28.8	28.6	24.8	18.6	11.1	2.3	(25)
(26)			MCWB	-2.7	0.1	4.5	10.8	15.4	20.0	24.1	23.9	19.5	14.4	8.2	0.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.8	4.5	9.7	15.6	19.6	23.5	27.8	27.2	24.3	19.2	13.1	6.2	(27)
(28)			MCDWB	3.4	7.0	14.4	20.0	24.8	27.4	30.1	29.7	27.1	22.1	15.6	7.6	(28)
(29)		2%	WB	0.0	2.8	7.9	13.9	18.3	22.4	26.6	26.4	22.8	17.7	11.5	3.6	(29)
(30)			MCDWB	1.8	5.2	11.8	18.6	23.2	26.3	29.5	29.0	25.7	20.3	14.0	5.3	(30)
(31)		5%	WB	-1.2	1.5	6.4	12.6	17.4	21.7	25.7	25.7	21.7	16.4	10.1	2.0	(31)
(32)			MCDWB	0.6	4.0	10.0	17.2	22.0	25.6	28.5	28.3	24.6	19.2	12.5	3.6	(32)
(33)		10%	WB	-2.6	0.2	5.0	11.4	16.5	21.1	24.9	25.0	20.7	15.2	8.3	0.4	(33)
(34)			MCDWB	-0.8	2.8	8.5	15.9	21.1	24.9	27.8	27.6	23.7	18.0	10.8	2.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.2	7.9	7.5	7.9	7.5	6.2	5.5	6.1	8.2	8.5	7.8	7.6	(35)
(36)			MCDBR	9.1	8.9	9.0	9.9	9.2	8.0	6.8	6.8	8.9	8.5	8.9	8.6	(36)
(37)			MCWBR	7.7	6.1	5.5	4.9	4.2	3.3	3.0	3.1	4.3	5.6	7.1	7.5	(37)
(38)		5% WB	MCDWB	8.8	8.0	8.1	8.4	7.5	6.3	5.5	5.1	6.9	7.6	8.4	8.4	(38)
(39)			MCWBR	7.8	6.2	5.6	5.0	4.1	3.3	2.9	3.1	4.7	6.0	7.3	7.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.391	0.466	0.514	0.614	0.649	0.741	0.729	0.637	0.566	0.573	0.479	0.411	(40)	
(41)		taud	2.005	1.835	1.793	1.652	1.631	1.525	1.585	1.744	1.828	1.737	1.890	2.003	(41)	
(42)		Ebn at Noon	715	706	726	684	674	614	616	661	675	595	606	650	(42)	
(43)		Edh at Noon	122	169	197	242	253	282	263	218	188	185	138	114	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.35	3.16	4.51	5.24	5.96	5.31	4.83	4.78	4.41	3.30	2.38	2.01	(44)	
(45)		RadStd	0.20	0.21	0.31	0.33	0.52	0.49	0.41	0.36	0.28	0.24	0.23	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	-1.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page